

Particulate Matter Monitoring & Health Effects

Issue 2026-08-05 | Entry window 5 August 2026 | 10 records in scope

10

RECORDS IN SCOPE
(23 rejected, 7 held)

5

SENSING RECORDS
(50% OF CORPUS)

4

SUBTOPIC CLUSTERS
REPRESENTED

6

TIER-A
RECORDS

1

EFFECT ESTIMATE
EXTRACTED

Signal of the day

Monitoring capacity is now measurably *anti*-correlated with need: in eastern Canada, situationally vulnerable populations had 54% higher odds of living in a PM_{2.5} monitoring desert after adjusting for the concentration itself.

Siu et al. (*Environ Res Health*) applied **locally adaptive kernel density estimation** to all PM_{2.5} monitors — regulatory and low-cost — across Ontario, Quebec and Atlantic Canada, 2020–2024, setting each bandwidth from the local variability of surface PM_{2.5} and its sensitivity to distinguishing an exceedance. That is the methodological move worth stealing: coverage is defined against the concentration field's *own* spatial structure rather than against a fixed radius, so “monitored” means “able to resolve an exceedance here”, not “within x km of a device”. In mixed-effects logistic regression, residential instability carried **25% higher odds** and situationally vulnerable populations **54% higher odds** of monitoring deprivation, *after* controlling for surface PM_{2.5} and emission-facility density — so this is not simply monitors following pollution. With Canada's annual standard tightening to $8 \mu\text{g m}^{-3}$ in 2030, the exceedance question becomes harder exactly where the network is thinnest. Read alongside Jafari et al. and the siting papers of 4 August, a coherent strand is forming: **siting is an optimisation problem with an objective function**, and the objective is not coverage area but resolvable exposure contrast in the populations that carry the burden.

What changed today

- **Entry window 5 August 2026, contiguous with the 4 August issue; no gap.** The health axis was the weakest this watch has recorded: PubMed returned 15 health-axis and 8 sensing-axis records for the 5 August entry date, of which **11 were rejected as off-topic** — diet scores, dementia checklists, a lead-exposure oncology paper, a correction notice, therapeutic aerosols, poultry bioaerosols. The signal came from the non-PubMed legs.
- **Half the corpus is sensing, and for the second consecutive issue that is network *design*, not calibration chemistry.** Siu et al. quantify where monitoring is absent; Jafari et al. solve fixed-and-mobile placement jointly as a constrained optimisation over a Rotterdam bus network, maximising spatiotemporal coverage subject to a budget *and* to a constraint that every mobile sensor passes a reference station often enough to stay calibrated. That last constraint is the interesting one: it makes recalibration cadence a term in the siting objective rather than a separate operational concern.
- **One held DOI from 4 August was recovered and shipped.** Jafari et al. (IEEE OJ-ITS) had no retrievable abstract in Crossref, Europe PMC or Consensus on 4 August; a Semantic Scholar Graph query returned it on 5 August. **Semantic Scholar is now a fourth abstract source in the retry chain** — it is the only index that resolved any of the five held records. The other four, all Elsevier, remain unretrievable after a second attempt and stay held rather than rejected.
- **Gäbel & Hertig answer the question the calibration literature keeps deferring: how often.** A yearlong Augsburg co-location across –10 to 36 °C and 19–96 % RH finds that a single pre-deployment calibration decays, and that **monthly in-season recalibration** is what keeps an SPS30 inside the EU/EPA indicative-measurement data-quality objective. Carried as an out-of-window backfill (*Atmos. Meas. Tech.*, created 19 February 2026) — AMT is structurally invisible to PubMed, which is exactly the gap the Consensus leg exists to close. It must be excluded from any weekly trend line.
- **Sea breeze as a monitoring-representativeness problem, not a meteorology result.** Yang et al. identify 428 diurnal sea-breeze events in a 10-year coastal record and show that excluding them changes the diurnal amplitude *and* the seasonal cycle of O₃ and CH₄ under southwesterly flow. A coastal background site is therefore not a background site on those days. Aerosol number showed a pronounced diurnal cycle attributed to new particle formation from accumulated ship SO₂ — a mechanism that generates ultrafine number with negligible mass, invisible to a mass-based network.
- **Only one effect estimate carries an interval, and it is from a preprint.** Zhang et al. (medRxiv, not peer reviewed) estimate 7.1% (95% UI 5.4–8.8%) of cardiorespiratory ED visits averted by combined EGU, mobile-source and port policies across NYC, LA and Atlanta. Fourth consecutive issue without a peer-reviewed PM ratio estimate with a confidence interval — see the forest-plot caption so a rollout does not read this as pipeline loss.

Corpus analytics

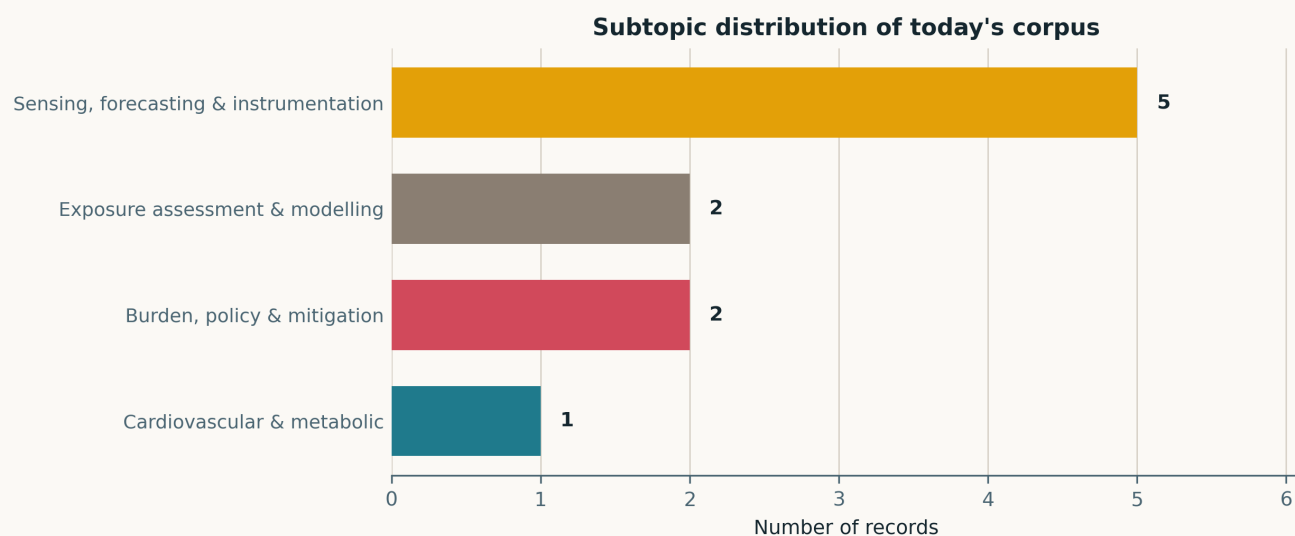


Fig. Subtopic assignment of the 10 in-scope records, single-label by dominant contribution. *Sensing, forecasting & instrumentation* takes 5 of 10 (50%) — the highest share this watch has recorded — ahead of exposure assessment and burden/policy at two each. **Six of the ten standing subtopics are empty**, including every mechanistic, respiratory and reproductive cluster: an unusually narrow window, driven by an exceptionally poor health-axis yield rather than by screening.

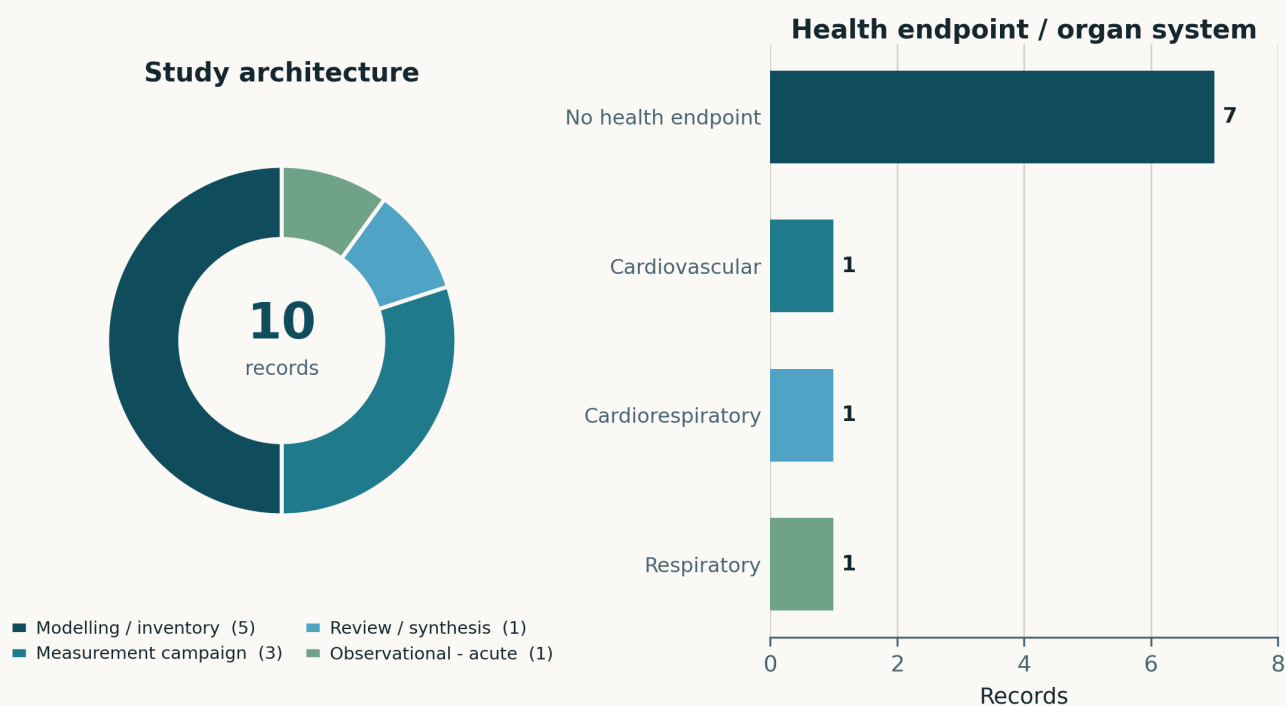


Fig. Left — study architecture. Modelling, inventory and geospatial work leads with 5/10, ahead of measurement campaigns at 3; there is **no experimental toxicology and no cohort epidemiology in this window at all**, and the single acute-observational entry is a preprint. Right — health endpoint by organ system. Seven of ten records carry no human health endpoint, consistent with a corpus that is half instrumentation. Note — the design map fell through to *Other / mixed* for six records on the first pass of this run, the fourth recurrence of that class; the cause was canonical group names being written directly onto records, and the map is now closed under its own output (see run log).

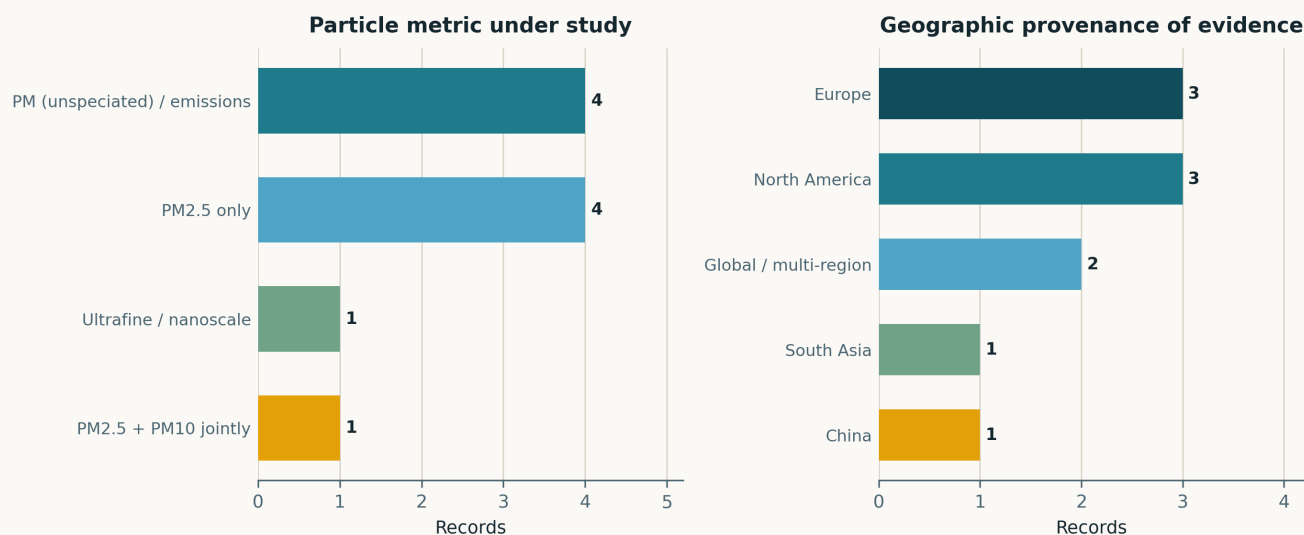


Fig. Left — particle metric. Four records use PM_{2.5} as the exposure; the rest are unspiciated PM, PM source factors from positive matrix factorisation, precursor-nitrogen emissions, or paired PM_{2.5}/PM₁₀/number. Right — geographic provenance. North America and Europe carry three each and China contributes only one — an inversion of the last two issues, where China supplied nearly half the corpus.

Life-course windows addressed today (records may span several stages)

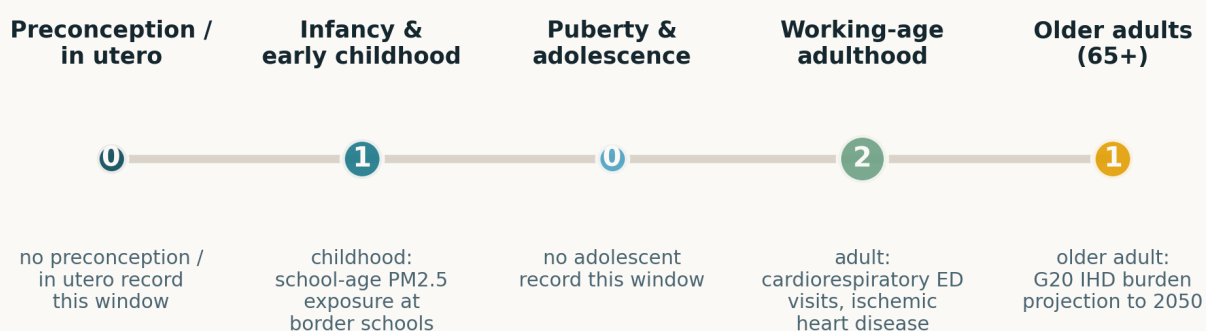


Fig. Life-course windows addressed; counts are non-exclusive. Only four of ten records carry a human life-course window at all. The childhood entry is the South Texas school deployment; the older-adult entry is a burden projection rather than a measured cohort, so the apparent coverage is thinner than the bar heights suggest.

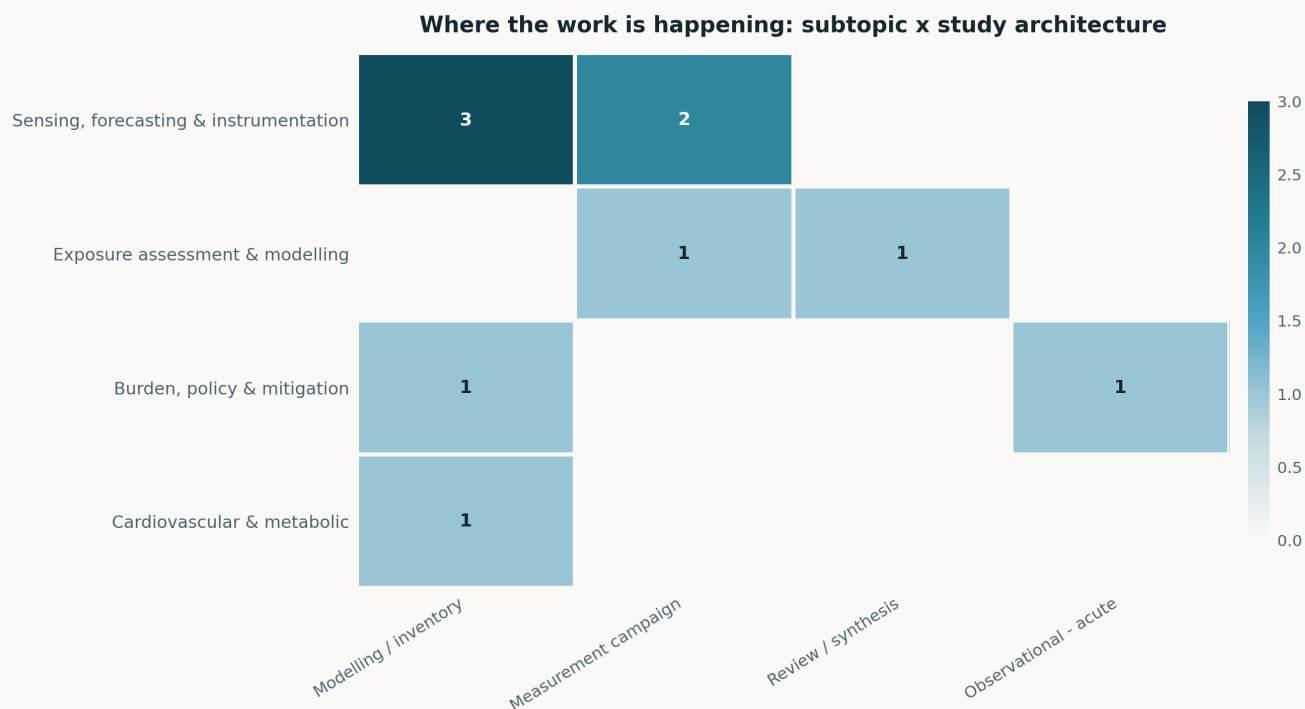


Fig. Subtopic $\times$ study-architecture cross-tabulation. The standing structural feature persists: the *Sensing* row contains no health-endpoint linkage. With 10 records across four subtopics the matrix is very sparse — read it as a coverage map, not a distribution.

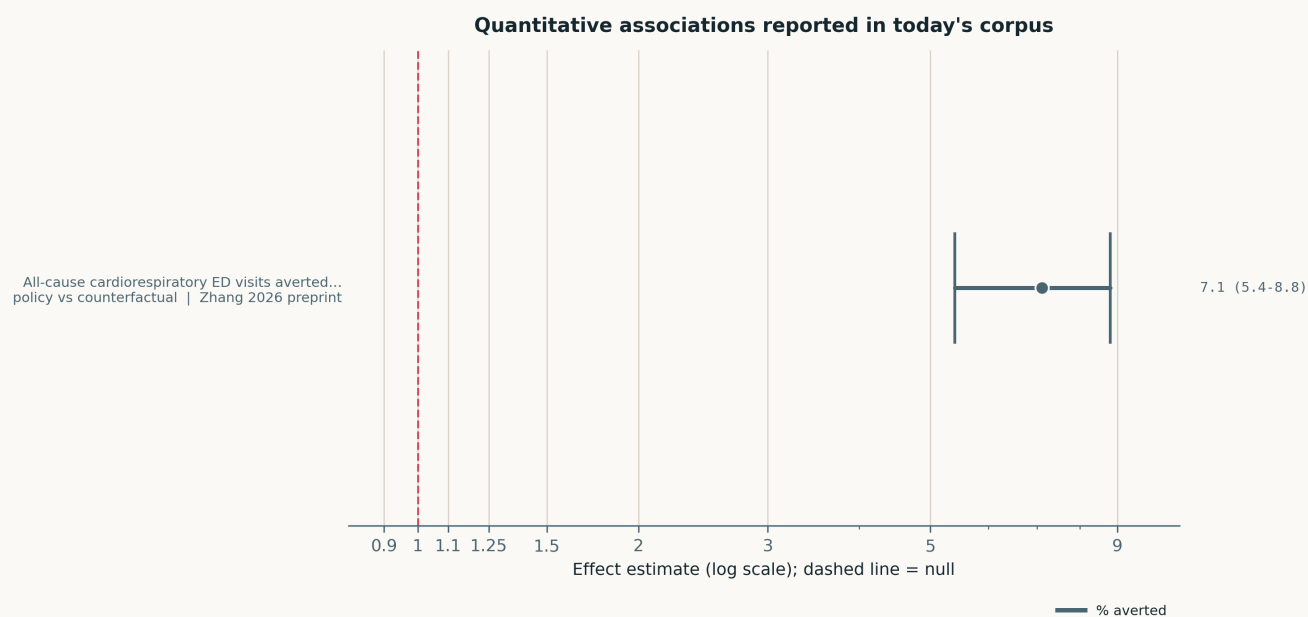


Fig. The single estimate plotted is a policy-attributable fraction from a preprint, not a peer-reviewed PM concentration–response estimate. Zhang et al. (medRxiv) report 7.1% (95% UI 5.4–8.8%) of cardiorespiratory ED visits averted by combined EGU, mobile-source and port policies across three US cities, 2005–2019, with median $PM_{2.5}$ reductions of 27–62%. No peer-reviewed record in this entry window reported a $PM_{2.5}$ or PM_{10} ratio estimate with a confidence interval — the fourth consecutive issue for which that holds. The plot is included so the absence is recorded explicitly rather than inferred from a missing figure.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

5 records

Siu et al. 2026 Environ Res Health

Disparities in fine particulate matter (PM_{2.5}) monitoring capacity in eastern Canada

Locally adaptive kernel density estimation of total PM_{2.5} monitor density — regulatory plus low-cost — across Ontario, Quebec and Atlantic Canada, 2020–2024, with each local bandwidth set from the surrounding variability of surface PM_{2.5} and its sensitivity to resolving an exceedance. Mixed-effects logistic regression gives **25% higher odds of monitoring deprivation for residential instability and 54% higher odds for situationally vulnerable populations**, after adjusting for surface PM_{2.5} and emission-facility density; suburban Montreal and Toronto, the Windsor and Niagara border regions, southeast Kitchener and remote northwest Ontario are named. Limitation: monitor density is a proxy for information yield, and low-cost and regulatory devices are pooled without weighting by data quality, so a dense but uncalibrated cluster counts as capacity. The adaptive-bandwidth construction is directly reusable for designing where to add nodes rather than merely auditing where they are.

doi: 10.1088/2752-5309/ae87c0

Jafari et al. 2026 IEEE Open J Intell Transp Syst

Joint optimization of bus-based mobile sensing and reference sensor deployment for urban air quality monitoring

Mathematical-programming formulation selecting simultaneously **which buses to instrument with low-cost sensors and where to place reference stations**, maximising spatiotemporal network coverage under a budget constraint while requiring that every mobile sensor pass a reference site frequently enough to be recalibrated. Demonstrated on a Rotterdam case study covering residential and industrial areas. The contribution is structural rather than empirical: recalibration cadence enters the siting objective as a constraint, so coverage and data quality are traded off in one problem instead of sequentially. Limitation: coverage is geometric — optimised against route and population geometry, with no validation against a measured concentration field, and no treatment of the sampling bias that fixed bus routes impose. This is the record held on 4 August and recovered on 5 August via Semantic Scholar.

doi: 10.1109/ojits.2026.3706855

Yang et al. 2026 Atmos Chem Phys

Diurnal sea breeze worsens coastal air quality and complicates monitoring of remote North Atlantic air

Ten years of observations from Penlee Point Atmospheric Observatory on the northeast Atlantic coast, from which **428 diurnal sea-breeze events** were identified, most frequent in spring and summer under low wind and strong irradiance. Daytime surface O₃, NO_x, CH₄, CO₂, SO₂, total aerosol number, PM_{2.5} and PM₁₀ were all elevated during events, with mean daytime O_x near **45 ppb** and increased regulatory exceedance for O₃. Accumulated ship-emitted SO₂ appears to drive new particle formation and a pronounced diurnal cycle in aerosol number. The monitoring payload: **excluding sea-breeze events reduces the diurnal amplitude and alters the seasonal cycle of O₃ and CH₄** under southwesterly flow — a coastal “background” site is conditionally not background. Limitation: one site, so the event classification and its consequences are not shown to transfer. Number-without-mass is precisely the regime a mass-based low-cost network cannot see.

doi: 10.5194/acp-26-10909-2026

Gäbel & Hertig 2026 Atmos Meas Tech

Recalibration of low-cost O₃ and PM_{2.5} sensors: linking practices to recent air sensor test protocols

Yearlong urban-background co-location at Augsburg of Alphasense OX-B431 and Sensirion SPS30 against reference instruments, spanning –10 to 36 °C, 19–96 % RH and PM_{2.5} to 153 µg m^{–3}, with MLR, ridge, random forest and XGBoost calibration functions evaluated against the current US EPA and CEN air-sensor test protocols. A five-month baseline training gives PM_{2.5} R² 0.93–0.97 in training and **0.84–0.93 in the following test period**, RMSE 2.04–3.02 µg m^{–3} — the decay is the finding. Testing pairwise recalibration cycles, **monthly in-season recalibration** was required to keep uncertainty inside the data-quality objective for indicative measurements. Limitation: one site, one year, two sensor models, so the cadence is an existence result rather than a general rule. **Out-of-window backfill** (created 19 Feb 2026); *AMT* is not indexed by PubMed. Exclude from weekly trend lines.

doi: 10.5194/amt-19-1293-2026

Wang et al. 2026 Environ Monit Assess

Traffic-induced environmental health assessment at intersections using roadside LiDAR

Roadside LiDAR extracts high-resolution vehicle trajectories at one urban and one suburban intersection, from which traffic emissions, noise and ground vibration are estimated and combined into a fuzzy comprehensive assessment score, with weights set by quadratic programming under ranking constraints, variable-weight grey entropy, and game theory to reconcile the two. The urban intersection scored **69.45 by day, falling to 57.12 at night (~18%)**, while the suburban site was stable at 72.80/74.54; population-dimension scores were consistently below environmental-dimension scores. Limitation: PM is *inferred* from trajectories through an emission model and never measured, so this is a modelled exposure surrogate validated against nothing — the elaborate weighting machinery sits on top of an unvalidated emissions estimate. Included because trajectory-derived emission proxies are an increasingly common input to near-road sensor fusion, and their weak point should be explicit.

doi: 10.1007/s10661-026-15750-y

Exposure assessment & modelling

2 records

Pinakana et al. 2026 NPJ Clean Air*Influence of transboundary wildfire smoke on fine particulate matter concentrations at South Texas border schools*

Low-cost sensors deployed at schools in Roma and Rio Grande City, Starr County, on the US–Mexico border for **146 days, November 2023 to April 2024**, corrected by collocation-based calibration against a research-grade instrument and read alongside meteorology, back trajectories and satellite smoke and fire products. Daily mean PM_{2.5} was significantly higher on smoke-impacted days ($p < 0.001$), with **site-level increases of 8.61% to 54.9%** — a sixfold spread across sites within one small county, which is itself the finding: transboundary smoke impact is not spatially uniform at the scale a single regulatory monitor would represent. Trajectories attribute the source to Mexican wildfire emissions. Limitation: one season, no indoor measurement despite the school framing, and attribution rests on trajectory plus satellite coincidence rather than on composition. A clean template for low-cost sensor deployment in an under-monitored community.

doi: 10.1038/s44407-026-00096-z

Phalen et al. 2026 Inhal Toxicol*Inhaled aerosol dosimetry update*

Synthesis of the Fourth International Aerosol Dosimetry Conference (Irvine, 2024) covering computational deposition models, in vitro and in vivo exposure systems, and deposition variability. Reported advances: integration of model types to span all major respiratory-tract regions; improved in vitro particle delivery and dose measurement; the upper airways recognised as a significant entry route. Two findings matter for ambient PM work — **dose variability emerged as the critical cross-cutting issue**, and **deposited ultrafine particles may acquire surface coatings after deposition and translocate into tissue, including fetal tissue**. Limitation: a conference report, not a systematic review — coverage reflects who presented, there are no effect sizes and no pooling, and the fetal-translocation claim is stated without the primary evidence. Useful mainly as a statement of where dosimetric uncertainty sits, which is the term that converts a measured ambient concentration into an internal dose.

doi: 10.1080/08958378.2026.2691193

Burden, policy & mitigation

2 records

Zhang et al. 2026 (preprint) medRxiv (preprint, not peer reviewed)*Health impacts of national and local air pollution control policies targeting electric generating units, mobile sources, and port activities in three US cities*

Counterfactual policy evaluation for New York City, Los Angeles and Atlanta, 2005–2019. Daily PM_{2.5}, criteria gases, 1-in-3-day PM_{2.5} components and PMF-resolved PM sources at six monitoring sites were used to construct the concentrations that would have occurred without the selected EGU, mobile-source and port policies; city-site-specific multi-pollutant Poisson models then linked observed concentrations to respiratory emergency-department visits and hospitalised cardiovascular ED visits, and the fitted models were applied to both concentration series. **Median PM_{2.5} reductions of 27–62% across cities, and 7.1% (95% UI 5.4–8.8%) of cardiorespiratory visits averted**. Limitation: not peer reviewed; the counterfactual is a modelled construct whose uncertainty is not propagated into the health interval, and multi-pollutant Poisson models over highly correlated criteria pollutants cannot cleanly assign the benefit to PM. The source-resolved PMF layer is the part worth following.

doi: 10.64898/2026.08.04.26359640

Goyal et al. 2026 Nat Commun*Quantifying environmental co-benefits of nitrogen-based crop restructuring and its implications on India's interstate trade network*

Integrated nutrient, water, greenhouse-gas, economic and interstate-trade accounting for Indian agriculture, comparing nitrogen-focused against water-focused crop-restructuring pathways. The nitrogen-focused pathway cuts **nitrogen surplus by 13.4%, water use by 18.6% and greenhouse gases by 8.7%** while holding baseline calorie production, and delivers about **4.6-fold larger reciprocal co-benefits** than the water-focused alternative, worth roughly **USD 1.19 bn** in avoided nitrogen-pollution socio-environmental cost. Limitation: PM is entirely implicit — reactive nitrogen is a dominant precursor of secondary inorganic aerosol over the Indo-Gangetic Plain, but no ambient PM_{2.5} response is modelled and no health endpoint is estimated, so the PM-relevant co-benefit is asserted through the nitrogen accounting rather than computed. Carried because agricultural NH₃ is the least-constrained term in South Asian PM source apportionment.

doi: 10.1038/s41467-026-75905-w

Cardiovascular & metabolic

1 record

Shi et al. 2026 Catheter Cardiovasc Interv*Ischemic heart disease in G20 countries: forecasted burden and regional disparities based on GBD 2023 estimates*

GBD 2023 descriptive burden analysis of ischemic heart disease incidence, mortality and DALYs across the G20, 1990–2023, with EAPC and joinpoint regression, population attributable fractions for modifiable risks, ARIMA and Bayesian age-period-cohort projection to 2050, and XGBoost with SHAP for DALY determinants. **8.52 M cases and 5.81 M deaths in 2023**; age-standardised incidence and mortality fell (EAPC –1.61 and –1.49) but incidence rose in women aged 15–49, with Russia highest on all three age-standardised rates and Japan projected to peak again around 2032. High systolic blood pressure dominated the risk attribution, with **particulate matter and lead named as the prominent environmental contributors**. Limitation: GBD population attributable fractions are modelled from the same exposure–response literature this watch tracks, so this is not independent evidence of a PM effect — it is that literature propagated through a burden model, and the machine-learning layer adds interpretability, not identification. Tier C: read as context, not as evidence.

doi: 10.1002/ccd.70783

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Shi et al.	Ischemic heart disease in G20 countries: forecasted burden and regional disparities based on GBD 2023 estimates	<i>Catheter Cardiovasc Interv</i>	Modelling / inventory	Unspeciated PM	Multi-country
Sensing, forecasting & instrumentation					
Gäbel & Hertig	Recalibration of low-cost O3 and PM _{2.5} sensors: linking practices to recent air sensor test protocols	<i>Atmos Meas Tech</i>	Field campaign / remote sensing	PM _{2.5}	Europe
Jafari et al.	Joint optimization of bus-based mobile sensing and reference sensor deployment for urban air quality monitoring	<i>IEEE Open J Intell Transp Syst</i>	Modelling / inventory	Unspeciated PM	Europe
Siu et al.	Disparities in fine particulate matter (PM _{2.5}) monitoring capacity in eastern Canada	<i>Environ Res Health</i>	Spatial analysis / GIS	PM _{2.5}	North America
Wang et al.	Traffic-induced environmental health assessment at intersections using roadside LiDAR	<i>Environ Monit Assess</i>	Modelling / inventory	Unspeciated PM	China
Yang et al.	Diurnal sea breeze worsens coastal air quality and complicates monitoring of remote North Atlantic air	<i>Atmos Chem Phys</i>	Field campaign / remote sensing	PM _{2.5} ; PM ₁₀ ; total aerosol number	Europe
Exposure assessment & modelling					
Phalen et al.	Inhaled aerosol dosimetry update	<i>Inhal Toxicol</i>	Review / synthesis	Unspeciated PM; ultrafine	Multi-country
Pinakana et al.	Influence of transboundary wildfire smoke on fine particulate matter concentrations at South Texas border schools	<i>NPJ Clean Air</i>	Field campaign / remote sensing	PM _{2.5}	North America
Burden, policy & mitigation					
Goyal et al.	Quantifying environmental co-benefits of nitrogen-based crop restructuring and its implications on India's interstate trade network	<i>Nat Commun</i>	Modelling / inventory	Precursor emissions (N)	India
Zhang et al. (preprint)	Health impacts of national and local air pollution control policies targeting electric generating units, mobile sources, and port activities in three US cities	<i>medRxiv (preprint)</i>	Time-series / case-crossover	PM _{2.5} ; PM sources (PMF)	North America

Provenance and method

Entry window **5 August 2026**, contiguous with the 4 August issue; no gap. Nine of the ten shipped records carry a 5 August entry or Crossref created date; the tenth (Gäbel & Hertig) is an explicitly flagged out-of-window backfill. **Sources.** PubMed E-utilities and the PubMed connector on two separate axes (health, instrumentation) windowed on [EDAT] — 15 health and 8 sensing hits after merging the two, of which 11 were rejected as off-topic. Europe PMC CREATION_DATE — 2 hits, 1 carried (a medRxiv preprint). Crossref scoped by journal ISSN across eight aerosol and atmospheric titles that PubMed does not index — 7 hits, 1 carried, 3 rejected as out of scope, 3 held for want of an abstract. Consensus on three sensing axes (calibration and drift, satellite-surface fusion, carbonaceous source apportionment) — every hit resolved to a DOI created before this window; **1 backfill carried, 9 rejected**. OpenAlex returned HTTP 429 and remains paid-blocked for date windowing; arXiv returned nothing in scope after the recency screen. **Screening.** Deduplication keys PMID → lowercased DOI → title signature, checked against `seen.json` before summarising; Consensus titles were resolved to DOIs through Crossref *before* screening, which is what prevented three re-summaries of records already shipped. **Seven records are held, not rejected:** their DOIs verify in Crossref but no abstract is retrievable from Crossref, Europe PMC, Semantic Scholar or Consensus. **Semantic Scholar was added to the retry chain on this run** and recovered one of the five records held on 4 August; the remaining four, all Elsevier, stay held. Held records are deliberately absent from `seen.json` so a later issue can carry them. Nothing was summarised from a title alone, and no DOI was inferred from a journal prefix. **Labels.** Subtopic, design, particle-metric and geography assignments are single-label judgements from title, abstract, keywords and MeSH terms, not a controlled vocabulary. The single effect estimate is transcribed verbatim from a preprint; see the forest-plot caption. All DOI links resolve to the publisher of record and were checked by the automated DOI gate before this issue compiled. Abstracts remain the copyright of their publishers.